Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	SPARK SURGE
Selected Domain	2
Selected SDG Track (1–17)	8
Project / Solution Name	LARIS: Quotation Tracker for Malaysian SMEs
Name (Leader)	HARRISH GOVIND A/L SETHULINGAM
Email & Phone (Leader)	harrishgovind09@gmail.com · [0125220471]
University Affiliation(s)	Universiti Tun Hussein Onn Malaysia (UTHM)
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysia has about 1.07 million micro, small and medium enterprises (DOSM Economic Census 2023). They contribute 39.7% of GDP and employ 48.7% of the workforce (DOSM MSME Performance 2025). Many are made-to-order trades such as signage, printing and 3D printing, where every job starts with a quotation.

These owners run quotations through chat apps, supplier price lists in PDFs and photos, and spreadsheets. Customer enquiries arrive as voice notes in Manglish. Supplier costs change, but the quote is built from a price remembered or copied by hand. The owner must also chase unanswered quotes. The results are slow replies, quotes priced on stale costs that erode margin, enquiries that go unanswered, and no record of what was promised.

The delay also hurts cash flow. Experian reports that average business payment delays in Malaysia were 61 days in 2025, and SMEs lag large corporates. A micro business has little buffer for lost margin or lost jobs, and no staff to spend hours a day on paperwork.

This matters for SDG 8. When small firms lose jobs to slow quoting and leak margin on pricing errors, they grow slower, formalise less and create fewer decent jobs. Reducing that administrative load helps them compete using the tools they already have, which are chat and a phone, instead of requiring an expensive ERP system.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

LARIS is a quotation tracker for Malaysian SMEs that works inside the chat apps owners and customers already use. It is deployed on Replit and currently runs on a Telegram bot, with a WhatsApp Business integration built and awaiting Meta setup.

A customer sends an enquiry as text or a Manglish voice note. ILMU speech-to-text transcribes it and the ILMU language model extracts the items and quantities as structured JSON. A rules engine, not the AI, calculates the price from the supplier cost book and applies a margin guard, so the model never sets a price. The owner approves the draft, and LARIS sends the quote and tracks it, with reminders for unanswered quotes.

Staff can photograph a shelf or price board. A Qwen2.5-VL vision model served with vLLM on a RunPod GPU, together with Tesseract OCR that cross-checks every price and SKU, turns the photo into catalog items for review. Customers can then ask "is this available?", and the bot answers from confirmed catalog items only. It never invents stock or prices. A follow-up such as "quote 50 kg" creates a draft quotation for the owner to confirm. Supplier PDFs, photos, XLSX, DOCX and CSV price lists are imported the same way, and quotes export to PDF, XLSX and CSV.

From the SDC workshops we use GenAI APIs (ILMU and RunPod), a PostgreSQL database with row-level tenant isolation, and Replit cloud deployment. People stay in control: confidence checks, owner approval, role permissions and an audit log.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

- Backend: Python FastAPI, PostgreSQL with row-level security (tenant isolation), APScheduler for reminders.

- Frontend: React (TypeScript) web app.

- Channels: Telegram Bot API (live) and WhatsApp Business Cloud API (built, not yet connected to Meta).

- AI: ILMU API for Manglish speech-to-text and structured extraction. Qwen2.5-VL-7B-Instruct served with vLLM on a RunPod GPU pod for catalog photos. Tesseract OCR cross-checks vision output.

- Files: PDF, XLSX, DOCX and CSV import and export.

- Hosting and CI: Replit deployment, GitHub Actions running about 500 automated tests.

Diagram (text version for the appendix): Customer or staff on Telegram/WhatsApp → webhook (FastAPI on Replit) → ILMU STT and extraction → rules pricing engine and PostgreSQL cost book → owner approval in the React app → quote PDF/XLSX/CSV sent back to chat. Catalog photos go → RunPod Qwen2.5-VL plus Tesseract → catalog review → availability and quote-request replies. I can draw it with the archify skill if you want a picture.

Sponsor APIs used: Yes. ILMU for speech-to-text and extraction, and RunPod for the GPU vision model (pod running vLLM with Qwen2.5-VL-7B). Replit hosts the whole app. Check your RunPod proof PDF matches the pod you actually used.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We applied design thinking in five steps. Empathise: we mapped the journey of a made-to-order owner from enquiry to sent quote and wrote interview questions on how they quote, where supplier prices live and who chases late replies. Define: the core problem is slow, unrecorded and stale-priced quotes. Ideate: we chose chat channels over a new app because owners already live in them, and rules for pricing so AI cannot set a price. Prototype: we built the working loop in small steps (enquiry, extraction, pricing, approval, tracking, catalog photos) and tested every step in automated CI on each commit, about 500 tests. Test: we will time first-time owners doing a quote manually and with LARIS, using synthetic data only.

For deployment, LARIS runs on a Replit deployment with a managed PostgreSQL database. It uses row-level security so one business cannot see another's data, signed licences, role permissions, rate limits and an audit log. The GPU model runs on RunPod and can be stopped when idle to control cost. By Phase 2 we will finish the WhatsApp connection, add per-size prices to catalog items, and run the timing tests with real owners.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 8.3: "Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises, including through access to financial services." Secondary: 8.2 (productivity through technological upgrading).
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Average business payment delay in Malaysia is 61 days (Experian State of Credit 2025, via RinggitPlus); SMEs lag large corporates. Manual quote time per quote depending on large the quotation is and some still using pen and paper.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Minutes from enquiry to sent quote, plus % of enquiries answered within 1 hour.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	50% fewer minutes per quote (target to confirm against the timing baseline) and 90% of enquiries receiving a first reply within 1 hour.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [X] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [X] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [X] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [X] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
ilmu-mini-v3.3	Model	Commercial API
ilmu-asr-v4.2	Model	Commercial API
Qwen2.5-VL-7B-Instruct	Model	Apache-2.0
Mesolitica Malaysian Whisper	Model	Not Stated
Tesseract 5 (English)	Model	Apache-2.0
Backend (Python 3.13): FastAPI/Starlette, uvicorn, httpx, python-multipart, APScheduler, PyYAML	Code	MIT, BSD-3-Clause, Apache-2.0.
Documents and images: pypdf, Pillow, pytesseract, openpyxl, defusedxml, reportlab.	Code	BSD, MIT, Apache-2.0, PSF
PostgreSQL 16	Code	PostgreSQL Licence
psycopg 3	Code	LGPL-3.0
cryptography	Code	Apache-2.0 or BSD-3-Clause

Frontend: TypeScript, React 19, Vite with @vitejs/plugin-react, Tailwind CSS 4	Code	Apache-2.0, MIT.
Development and test only (not shipped): pytest, PGlite, axe-core	Code	MIT, Apache-2.0, MPL-2.0.

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team’s knowledge and that the project will be original work created during the SDC 2026 hackathon period.

HARRISH.

24 OCTOBER 2026.